2018-Fall Similar for 2022-Fall

8.(b) Find the DFT of the sequence using radix-2 DITFFT.

$$x(n) = \cos\frac{\pi}{2}n \; ; \quad 0 \le n \le 7$$

Sol Here,

$$x(n) = \cos\frac{\pi}{2}n \; ; \quad 0 \le n \le 7$$

Now,

n	$x(n) = \cos\frac{\pi}{2}n$
0	$x(0) = 1$
1	$x(1) = 0$
2	$x(2) = -1$
3	$x(3) = 0$
4	$x(4) = 1$
5	$x(5) = 0$
6	$x(6) = -1$
7	$x(7) = 0$

$$\therefore W_N^k = e^{-j2\pi k/N}$$

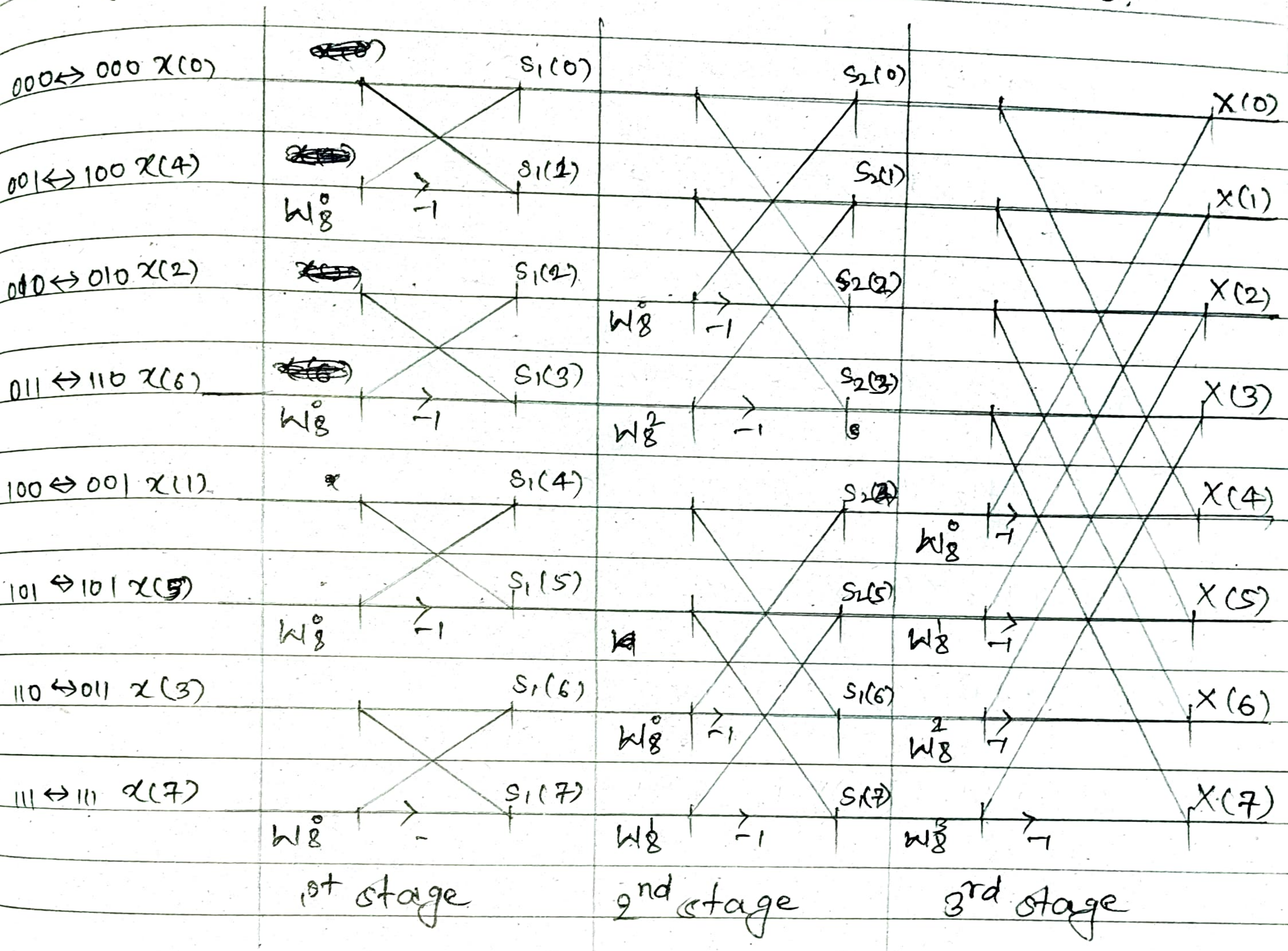
$$W_8^0 = 1$$

$$W_8^1 = 0.707 - j0.707$$

$$W_8^2 = -j$$

$$W_8^3 = -0.707 - j0.707$$

The graph of 8-point DIT FFT is shown below:



(i) Output for 1^{st} stage:

$S_1(0) = x(0) + W_8^0\, x(4) = 1 + 1 \times 1 = 2$

$S_1(1) = x(0) + W_8^0\, x(4) \times -1 = 1 + 1 \times 1 \times -1 = 0$

$S_1(2) = x(2) + W_8^0\, x(6) = -1 + 1 \times -1 = -2$

$S_1(3) = x(2) + W_8^0\, x(6) \times -1 = -1 + 1 \times -1 \times -1 = 0$

$S_1(4) = x(1) + W_8^0\, x(5) = 0 + 1 \times 0 = 0$

$S_1(5) = x(1) + W_8^0\, x(5) \times -1 = 0 + 1 \times 0 \times -1 = 0$

$S_1(6) = x(3) + W_8^0\, x(7) = 0 + 1 \times 0 = 0$

$S_1(7) = x(3) + W_8^0\, x(7) \times -1 = 0 + 1 \times 0 \times -1 = 0$

Discrete Filter Structure
5.1 FIR filter overview
5.2 Structures for FIR Filter (direct, cascade, freq... lattice, lattice ladder...
...les, and zeros
5.3 II...

(ii) Output for 2nd stage:

$s_2(0) = s_1(0) + W_8^0 \, s_1(2) = 2 + 1 \times -2 = 0$

$s_2(1) = s_1(1) + W_8^2 \, s_1(3) = 0 + -j \times 0 = 0$

$s_2(2) = s_1(0) + W_8^0 \, s_1(2) \times -1 = 2 + 1 \times -2 \times -1 = 4$

$s_2(3) = s_1(1) + W_8^2 \, s_1(3) \times -1 = 0 + -j \times 0 \times -1 = 0$

$s_2(4) = s_1(4) + s_1(6) \, W_8^0 = 0 + 1 \times 0 = 0$

$s_2(5) = s_1(5) + W_8^2 \, s_1(7) = 0 + -j \times 0 = 0$

$s_2(6) = s_1(4) + W_8^0 \, s_1(6) \times -1 = 0 + 1 \times 0 \times -1 = 0$

$s_2(7) = s_1(5) + W_8^2 \, s_1(7) \times -1 = 0 + -j \times 0 \times -1 = 0$

$S_2(2)=4$

(iii) Output for 3rd stage:

$X(0) = S_2(0) + W_8^0 S_2(4) = 0 + 1 \times 0 = 0$

$X(1) = S_2(1) + W_8^1 S_2(5) = 0$

$X(2) = S_2(2) + W_8^2 S_2(6) = 4$

$X(3) = S_2(3) + W_8^3 S_2(7) = 0$

$X(4) = S_2(0) + W_8^0 S_2(4) \times -1 = 0$

$X(5) = S_2(1) + W_8^1 S_2(5) \times -1 = 0$

$X(6) = S_2(2) + W_8^2 S_2(6) \times -1 = 4$

$X(7) = S_2(3) + W_8^3 S_2(7) \times -1 = 0$

$\therefore X(K) = \{0, 0, 4, 0, 0, 0, 4, 0\}$ Ans

2021-Spring

3.a) Use DIF-FFT algorithm to compute 8-point DFT of $x(n) = \{2, 1, 1, 1\}$. (8)

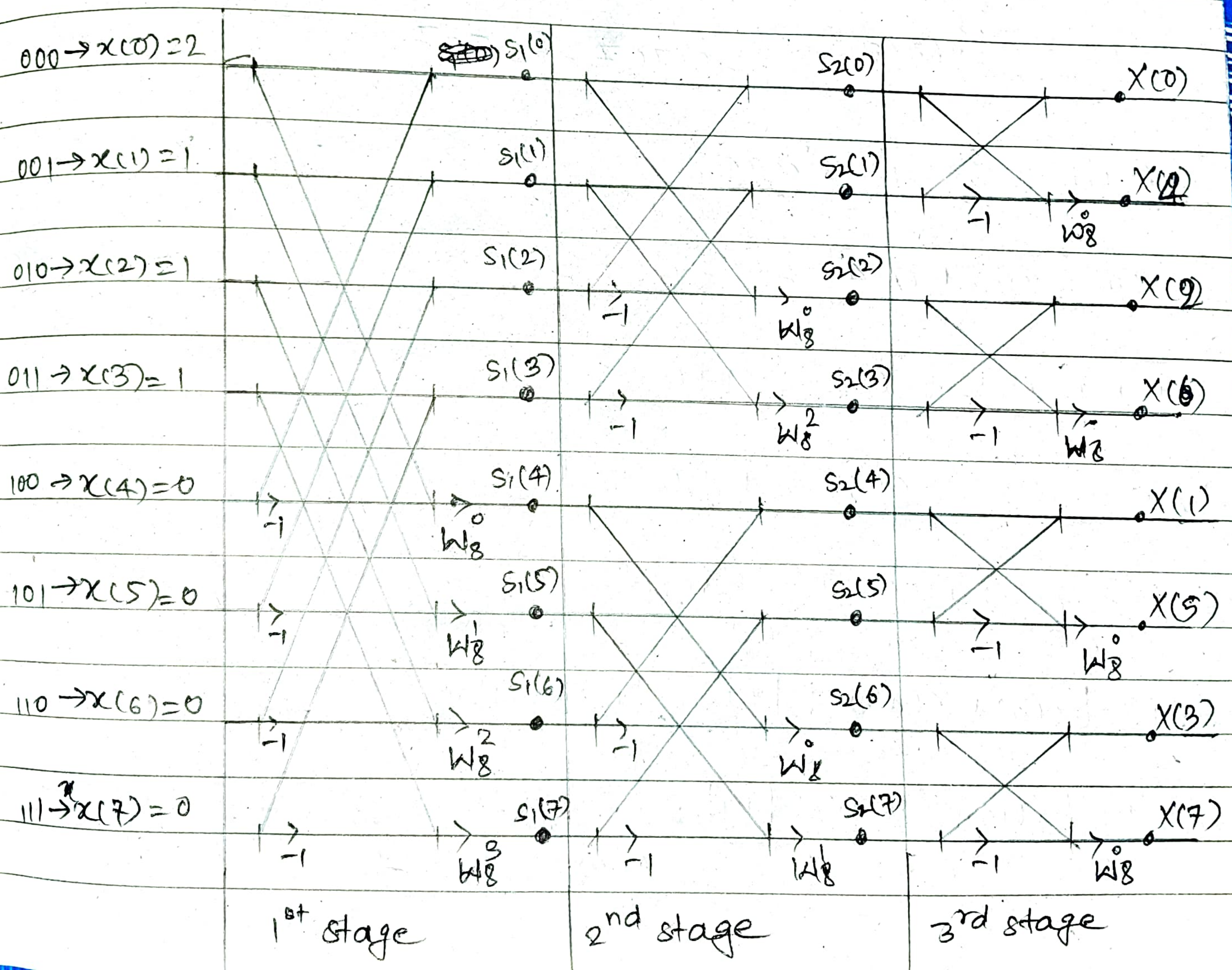
Sol: Here,

$x(n) = \{2, 1, 1, 1\}$

Since there is 8-point. So, $x(n)$ will be made as

$x(n) = \{2, 1, 1, 1, 0, 0, 0, 0\}$

The Graph of DIF-FFT is shown below:



Where,

$$W_N^k = e^{-j2\pi k/8}$$

$$W_8^0 = e^0 = 1$$

$$W_8^1 = e^{-j2\pi/8} = 0.707 - j0.707$$

$$W_8^2 = e^{-j4\pi/8} = -j$$

$$W_8^3 = e^{-j6\pi/8} = -0.707 - j0.707$$

(i) Output for 1st stage:

$$s_1(0) = x(0) + x(4) = 2 + 0 = 2$$

$$s_1(1) = x(1) + x(5) = 1 + 0 = 1$$

$$s_1(2) = x(2) + x(6) = 1 + 0 = 1$$

$$s_1(3) = x(3) + x(7) = 1 + 0 = 1$$

$$s_1(4) = [x(0) + x(4) \times -1]\, W_8^0 = (2 + 0 \times -1) \times 1 = 2$$

$$s_1(5) = [x(1) + x(5) \times -1]\, W_8^1 = (1 + 0 \times -1) \times (0.707 - j0.707)$$
$$= 0.707 - j0.707$$

.2 Structures for FIR
5.3 IIR

$$s_1(6) = [x(2) + x(6) \times -1]\, W_8^2 = (1 + 0 \times -1) \times -j = -j$$

$$s_1(7) = [x(3) + x(7) \times -1]\, W_8^3 = (1 + 0 \times -1)(-0.707 - j0.707)$$

$$= -0.707 - j0.707$$

(ii) Output for 2nd stage:

$$s_2(0) = s_1(0) + s_1(2) = 2 + 1 = 3$$

$$s_2(1) = s_1(1) + s_1(3) = 1 + 1 = 2$$

$$s_2(2) = [s_1(0) + s_1(2) \times -1]\, W_8^0 = (2 + 1 \times -1)\, 1 = 1$$

$$s_2(3) = [s_1(1) + s_1(3) \times -1]\, W_8^2 = (1 + 1 \times -1)(-j) = 0$$

$$s_2(4) = s_1(4) + s_1(6) = 2 + -j = 2 - j$$

$$s_2(5) = s_1(5) + s_1(7) = 0.707 - j0.707 - 0.707 - j0.707 = -j1.414$$

$$s_1(6) = [s_1(4) + s_1(6) \times -1]\, W_8^0 = [2 + (-j) \times -1]\, 1 = 2 + j$$

$$s_1(7) = [s_1(5) + s_1(7) \times -1]\, W_8^2 = [0.707 - j0.707 + (-0.707 - j0.707) \times -1](-j)$$

$$= \cancel{0.707}\ j1.414$$

(iii) Output for 3rd or last stage:

$X(0) = S_2(0) + S_2(1) = 3 + 2 = 5$

$X(4) = [S_2(0) + S_2(1) \times -1]\, W_8^0 = (3 + 2\times -1)\times 1 = 1$

$X(2) = [S_2(2) + S_2(3)] = 1 + 0 = 1$

$X(6) = [S_2(2) + S_2(3)\times -1]\, W_8^0 = (1 + 0\times -1)\times 1 = 1$

$X(1) = [S_2(4) + S_2(5)] = 2 - j + -j1.414 = 2 - j2.414$

$X(5) = [S_2(4) + S_2(5)\times -1]\, W_8^0 = [2 - j + -j1.414 \times -1]\times 1 = 2 + j0.414$

$X(3) = [S_2(6) + S_2(7)] = (2+j) + -j1.414 = 2 - j0.414$

$X(7) = [S_2(6) + S_2(7)\times -1]\, W_8^0 = [(2+j) + (-j1.414)\times -1]\times 1$
$= 2 + j2.414$

Hence,

$\cancel{X(K) = \{5, 1, 1, 1, 2-j2.414, 2+j0.414, 2-j0.414, 2+j2.414\}}$ ~~Ans~~

$X(K) = \{5, 2-j2.414, 1, 2-j0.414, 1, 2+j0.414, 1, 2+j2.414\}$ Ans